

A COUNTEREXAMPLE TO THE GALLIAN–SCHOENHARD STAR-FOREST CONJECTURE

ABSTRACT. Gallian and Schoenhard conjectured that a disjoint union of stars is strongly even harmonious whenever one component has more than two edges. We disprove the conjecture: $S_3 \sqcup 3S_1$ is properly even harmonious but not strongly even harmonious. The nonexistence holds even with vertex labels in the larger codomain $\{0, \dots, 2q\}$, and therefore also under the corrected proper codomain $\{0, \dots, 2q - 1\}$. A counting inequality also gives two infinite families of counterexamples.

1. THE COUNTEREXAMPLE

Write $S_n = K_{1,n}$, so S_n has n edges, and let $\sqcup$ denote disjoint union. In the original even-harmonious convention, a graph G with q edges is strongly even harmonious if there is an injection

$$f : V(G) \longrightarrow \{0, 1, \dots, 2q\}$$

such that the induced edge labels

$$f^*(uv) = f(u) + f(v) \pmod{2q}$$

are exactly $0, 2, \dots, 2q - 2$, and every edge uv satisfies

$$0 < f(u) + f(v) \leq 2q.$$

This is the convention used in the original paper and retained by Henderson [1, 3]. Gallian’s current survey instead builds the strong condition on the corrected properly even harmonious codomain $\{0, 1, \dots, 2q - 1\}$ [2, Section 4.6]; see also Henderson [3]. Since this is the smaller codomain, a nonexistence result for $\{0, \dots, 2q\}$ applies under both conventions.

Gallian and Schoenhard conjectured that

$$S_{n_1} \sqcup \dots \sqcup S_{n_t}$$

is strongly even harmonious whenever at least one n_i exceeds 2 [1, Conjecture 4.12]. The statement is still listed as a conjecture in the 2025 edition of Gallian’s survey [2, Section 4.6, p. 142].

Theorem 1. *The star forest*

$$G = S_3 \sqcup 3S_1$$

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is properly even harmonious but has no strongly even harmonious labeling, even with vertex labels in $\{0, 1, \dots, 12\}$. Consequently, Conjecture 4.12 of Gallian and Schoenhard is false under either codomain convention.

The proof uses the following counting obstruction.

Lemma 2. *Let*

$$F = S_{n_1} \sqcup \dots \sqcup S_{n_t}, \quad n_1 \geq \dots \geq n_t \geq 1, \quad q = \sum_{i=1}^t n_i.$$

If F has a strongly even harmonious labeling with vertex labels in $\{0, 1, \dots, 2q\}$, then

$$(1) \quad q^2 + \sum_{i=1}^t (i-1)(n_i-1) \leq \frac{(q-t+1)(3q+t)}{2}.$$

Proof. Let c_i be the center of S_{n_i} , and let U be the sum of the unused vertex labels. The strong bound lifts the edge residues $0, 2, \dots, 2q-2$ uniquely to the ordinary edge sums

$$2, 4, \dots, 2q,$$

whose total is $q(q+1)$. Therefore

$$q(q+1) = \sum_{v \in V(F)} f(v) + \sum_{i=1}^t (n_i-1)f(c_i).$$

Since the sum of all labels in $\{0, \dots, 2q\}$ is $q(2q+1)$,

$$(2) \quad U = q^2 + \sum_{i=1}^t (n_i-1)f(c_i).$$

Put $w_i = n_i - 1$, so $w_1 \geq \dots \geq w_t \geq 0$, and let $b_1 < \dots < b_t$ be the center labels in increasing order. Then $b_i \geq i-1$, and the rearrangement inequality gives

$$\sum_{i=1}^t (n_i-1)f(c_i) \geq \sum_{i=1}^t w_i b_i \geq \sum_{i=1}^t (i-1)(n_i-1).$$

There are $q-t+1$ unused labels. Their sum is at most the sum of the $q-t+1$ largest labels in the codomain, namely

$$U \leq (q+t) + (q+t+1) + \dots + 2q = \frac{(q-t+1)(3q+t)}{2}.$$

Combining this bound with (2) proves (1). $\square$

Proof of Theorem 1. For $G = S_3 \sqcup 3S_1$, we have $q = 6$, $t = 4$, and

$$(n_1, n_2, n_3, n_4) = (3, 1, 1, 1).$$

Thus the two sides of (1) are 36 and 33, respectively. Hence no strongly even harmonious labeling exists, even in the larger codomain $\{0, \dots, 12\}$.

It remains to exhibit a properly even harmonious labeling. Let c be the center of S_3 , with leaves x_1, x_2, x_3 , and let $a_i b_i$, $1 \leq i \leq 3$, be the three S_1 components. Set

$$f(c) = 0, \quad (f(x_1), f(x_2), f(x_3)) = (2, 6, 10),$$

and

$$(f(a_1), f(b_1)) = (3, 5), \quad (f(a_2), f(b_2)) = (4, 8), \quad (f(a_3), f(b_3)) = (7, 9).$$

These are distinct labels in $\{0, \dots, 11\}$, and the induced edge labels modulo 12 are exactly

$$0, 2, 4, 6, 8, 10.$$

Thus G is properly even harmonious. $\square$

Corollary 3. *Neither of the following star forests is strongly even harmonious under either codomain convention:*

$$S_3 \sqcup mS_1 \quad (m \geq 3), \quad S_3 \sqcup kS_2 \quad (k \geq 5).$$

Proof. For $S_3 \sqcup mS_1$, the left side of (1) exceeds the right side by

$$m^2 - 6 > 0.$$

For $S_3 \sqcup kS_2$, the weighted term in (1) is $k(k+1)/2$, and the corresponding difference is

$$k^2 - 3k - 6 > 0.$$

$\square$

Remark 4. Henderson proved that a union of at least two stars, each having at least two edges, is properly even harmonious [3, Theorem 3.1]. Consequently, $S_3 \sqcup kS_2$ is properly even harmonious but not strongly even harmonious for every $k \geq 5$.

REFERENCES

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